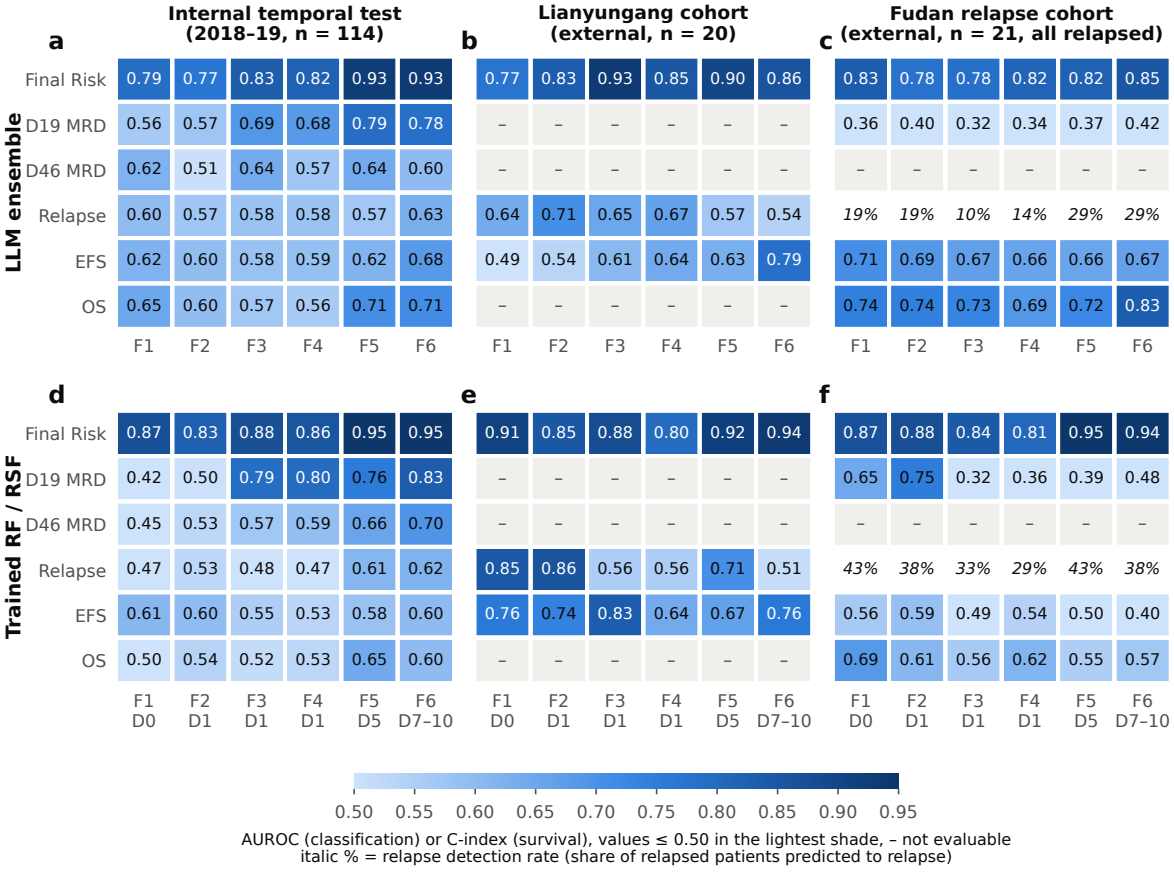
a
Internal temporal test (2018–19, n = 114)
b
Lianyungang cohort (external, n = 20)
c
Fudan relapse cohort (external, n = 21, all relapsed)
LLM ensemble
Final Risk 0.79 0.77 0.83 0.82 0.93 0.93 | 0.77 0.83 0.93 0.85 0.90 0.86 | 0.83 0.78 0.78 0.82 0.82 0.85
D19 MRD 0.56 0.57 0.69 0.68 0.79 0.78 | – – – – – – | 0.36 0.40 0.32 0.34 0.37 0.42
D46 MRD 0.62 0.51 0.64 0.57 0.64 0.60 | – – – – – – | – – – – – –
Relapse 0.60 0.57 0.58 0.58 0.57 0.63 | 0.64 0.71 0.65 0.67 0.57 0.54 | 19% 19% 10% 14% 29% 29%
EFS 0.62 0.60 0.58 0.59 0.62 0.68 | 0.49 0.54 0.61 0.64 0.63 0.79 | 0.71 0.69 0.67 0.66 0.66 0.67
OS 0.65 0.60 0.57 0.56 0.71 0.71 | – – – – – – | 0.74 0.74 0.73 0.69 0.72 0.83
F1 F2 F3 F4 F5 F6 | F1 F2 F3 F4 F5 F6 | F1 F2 F3 F4 F5 F6
d
e
f
Trained RF / RSF
Final Risk 0.87 0.83 0.88 0.86 0.95 0.95 | 0.91 0.85 0.88 0.80 0.92 0.94 | 0.87 0.88 0.84 0.81 0.95 0.94
D19 MRD 0.42 0.50 0.79 0.80 0.76 0.83 | – – – – – – | 0.65 0.75 0.32 0.36 0.39 0.48
D46 MRD 0.45 0.53 0.57 0.59 0.66 0.70 | – – – – – – | – – – – – –
Relapse 0.47 0.53 0.48 0.47 0.61 0.62 | 0.85 0.86 0.56 0.56 0.71 0.51 | 43% 38% 33% 29% 43% 38%
EFS 0.61 0.60 0.55 0.53 0.58 0.60 | 0.76 0.74 0.83 0.64 0.67 0.76 | 0.56 0.59 0.49 0.54 0.50 0.40
OS 0.50 0.54 0.52 0.53 0.65 0.60 | – – – – – – | 0.69 0.61 0.56 0.62 0.55 0.57
F1 D0 F2 D1 F3 D1 F4 D1 F5 D5 F6 D7–10 | F1 D0 F2 D1 F3 D1 F4 D1 F5 D5 F6 D7–10 | F1 D0 F2 D1 F3 D1 F4 D1 F5 D5 F6 D7–10
0.50 0.55 0.60 0.65 0.70 0.75 0.80 0.85 0.90 0.95
AUROC (classification) or C-index (survival), values ≤ 0.50 in the lightest shade, – not evaluable
italic % = relapse detection rate (share of relapsed patients predicted to relapse)